

Chapter 13

Polyfunctional Acids

Fundamentals of Analytical Chemistry, Skoog, et al., 10th Ed., 2022

Chapter 13 Problems:

13-5ce, 13-10ace, *13-11ace, 13-12ace, 13-14, 13-22, 13-34ace

*Use expanded form for amphotere

Polyfunctional Acids

Polyfunctional acids are weak acids that have more than one proton capable of being donated.



k_{a1} = equilibrium constant for a polyfunctional acid giving up the first H^+

$$k_{a1} = \frac{[\text{H}_3\text{O}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = 4.2 \times 10^{-7}$$

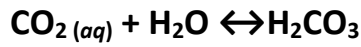


k_{a2} = equilibrium constant for a polyfunctional acid giving up the second H^+

$$k_{a2} = \frac{[\text{H}_3\text{O}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]} = 5.62 \times 10^{-11}$$

$k_{a1} \gg \gg \gg k_{a2}$ (10,000 times greater)

Carbon Dioxide (CO₂) System

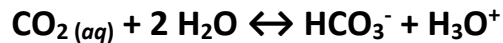
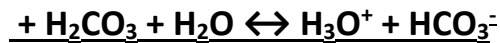
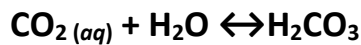


$$k_{\text{hydration}} = \frac{[\text{H}_2\text{CO}_3]}{[\text{CO}_2]} = 2.8 \times 10^{-3}$$



$$k_1 = \frac{[\text{H}_3\text{O}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = 1.5 \times 10^{-4}$$

Taking the sum of the two reactions above



$$k_{a1} = k_{\text{hydration}} * k_1 = (2.8 \times 10^{-3}) * (1.5 \times 10^{-4}) = 4.2 \times 10^{-7}$$



$$k_{a2} = \frac{[\text{H}_3\text{O}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]} = 5.62 \times 10^{-11}$$

1. Calculate the pH of a 0.100 M solution of H_2CO_3

2. Calculate the pH of a 0.130 M solution of Na_2CO_3

Amphiprotic Species



$$k_{a2} = \frac{[H_3O^+][A^{2-}]}{[HA^-]}$$



$$k_{b2} = \frac{k_w}{k_{a1}} = \frac{[H_2A][OH^-]}{[HA^-]}$$

If $k_{b2} > K_{a2} \rightarrow$ basic solution

If $k_{b2} < K_{a2} \rightarrow$ acidic solution

$$[H_3O^+] = \sqrt{\frac{k_{a2}C_{NaHA} + k_w}{1 + C_{NaHA}/k_{a1}}}$$

Where C_{NaHA} = concentration of the amphoteric species

(use the above equation to solve homework problem 13-11)

Where C_{NaHA}/k_{a1} is larger than 1 and $k_{a2}C_{NaHA}$ is much larger than k_w

the equation above simplifies to

$$[H_3O^+] = \sqrt{k_{a1}k_{a2}}$$

3. Calculate the pH of a 0.1000 M solution of NaHCO_3 .

4. Calculate the pH when 11.0 mL of 0.125 M sodium carbonate is mixed with 10.0 mL of 0.110 M hydrochloric acid.

The Phosphoric Acid System



k_{a1} = equilibrium constant for a polyfunctional acid giving up the first H^+

$$ka_1 = \frac{[\text{H}_3\text{O}^+][\text{H}_2\text{PO}_4^-]}{[\text{H}_3\text{PO}_4]} = 7.11 \times 10^{-3}$$



k_{a2} = equilibrium constant for a polyfunctional acid giving up the second H^+

$$ka_2 = \frac{[\text{H}_3\text{O}^+][\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]} = 6.32 \times 10^{-8}$$



k_{a3} = equilibrium constant for a polyfunctional acid giving up the third H^+

$$ka_3 = \frac{[\text{H}_3\text{O}^+][\text{PO}_4^{3-}]}{[\text{HPO}_4^{2-}]} = 4.5 \times 10^{-13}$$

$$ka_1 \gg ka_2$$

Why?

- 1) Due to electrostatic forces (*2nd H^+ is harder to remove from doubly charged anion*).
- 2) Statistical reason (*In the removal of the 1st proton there are more locations, than in the 2nd step*)

Phosphate Buffer System

H_3PO_4	weak acid (must use quadratic)
H_2PO_4^-	amphotere
HPO_4^{2-}	amphotere
PO_4^{3-}	conjugate base (must use quadratic) <i>Fully deprotonated conjugate base of a triprotic acid</i>

Buffers of a Polyfunctional Weak Acid

A buffer of a polyfunctional system can only exist between two species next to each other in a table of all species of the acid.

5. Calculate the pH of a solution that is 2.00 M solution of H_3PO_4 and 1.50 M KH_2PO_4 .

$$K_{a1} = 7.11 \times 10^{-3}$$

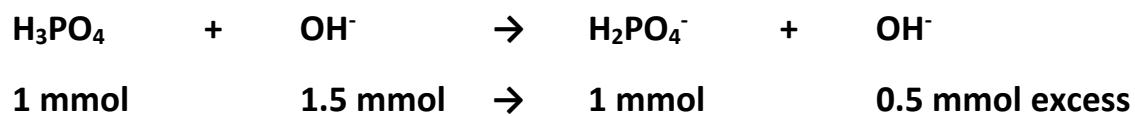
6. Calculate the pH of mixture of 20.00 mL of 0.113 M KH_2PO_4 and 30.00 mL 0.107 M Na_2HPO_4 are mixed.

H_3PO_4 $\text{pK}_{\text{a}1} = 2.15$

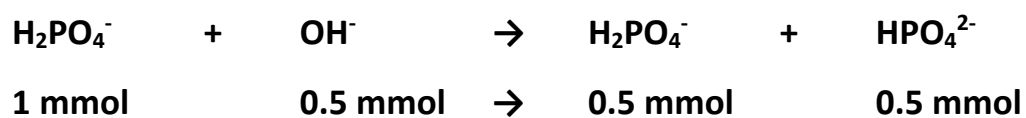
$\text{pK}_{\text{a}2} = 7.20$

$\text{pK}_{\text{a}3} = 12.30$

Addition of 1.5 mmol of hydroxide to 1 mmol of phosphoric acid



The reaction continues



$$\text{pH} = \text{pK}_{\text{a}2} + \log [\text{HPO}_4^{2-}] / [\text{H}_2\text{PO}_4^-]$$

$$\text{pH} = \text{pK}_{\text{a}2} + \log [0.5 \text{ mmol}] / [0.5 \text{ mmol}]$$

$$\text{pH} = \text{pK}_{\text{a}2}$$

7. Calculate the pH of solution when 12.00 mL of 0.113 M Na_3PO_4 and 11.00 mL 0.0800 M H_3PO_4 are mixed.

H_3PO_4 $\text{pK}_{\text{a}1} = 2.15$

$\text{pK}_{\text{a}2} = 7.20$

$\text{pK}_{\text{a}3} = 12.30$

Alpha Functions (α)

Give the fraction of each component in a polyfunctional acid system



α_0 = Fraction of fully protonated weak acid

$$\alpha_0 = \frac{[\text{H}_2\text{CO}_3]}{[\text{H}_2\text{CO}_3] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}]} = \frac{[\text{H}_2\text{CO}_3]}{C_T}$$

C_T = sum of the carbonate species

α_1 = Fraction that is missing one proton

$$\alpha_1 = \frac{[\text{HCO}_3^-]}{C_T}$$

α_2 = Fraction that is missing two protons

$$\alpha_2 = \frac{[\text{CO}_3^{2-}]}{C_T}$$

$$C_T = [H_2M] + [HM^-] + [M^{2-}]$$

$$\alpha_0 = \frac{[H_2M]}{C_T}$$

$$\alpha_1 = \frac{[HM^-]}{C_T}$$

$$\alpha_2 = \frac{[M^{2-}]}{C_T}$$

$$\alpha_0 + \alpha_1 + \alpha_2 = 1$$

$$\alpha_0 = \frac{[H_3O^+]^2}{[H_3O^+]^2 + k_{a1}[H_3O^+] + k_{a1}k_{a2}}$$

$$\alpha_1 = \frac{k_{a1}[H_3O^+]}{[H_3O^+]^2 + k_{a1}[H_3O^+] + k_{a1}k_{a2}}$$

$$\alpha_2 = \frac{k_{a1}k_{a2}}{[H_3O^+]^2 + k_{a1}[H_3O^+] + k_{a1}k_{a2}}$$

General Expression for H_nA

Denominator = D

$$D = [H_3O^+]^n + k_{a1}[H_3O^+]^{n-1} + k_{a1}k_{a2}[H_3O^+]^{n-2} + \dots k_{a1}k_{a2} \dots k_n$$

Numerator

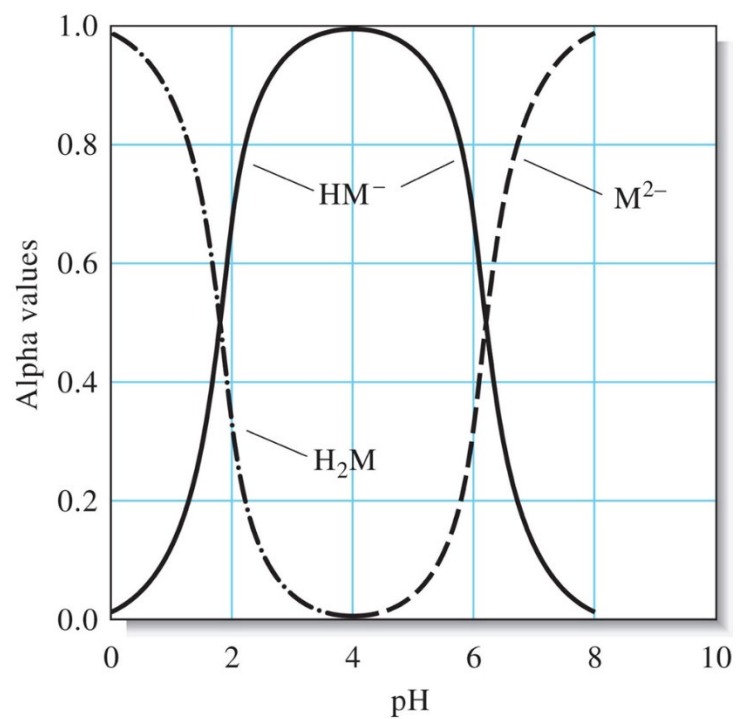
α_0 = 1st term from the denominator

α_1 = 2nd term from the denominator

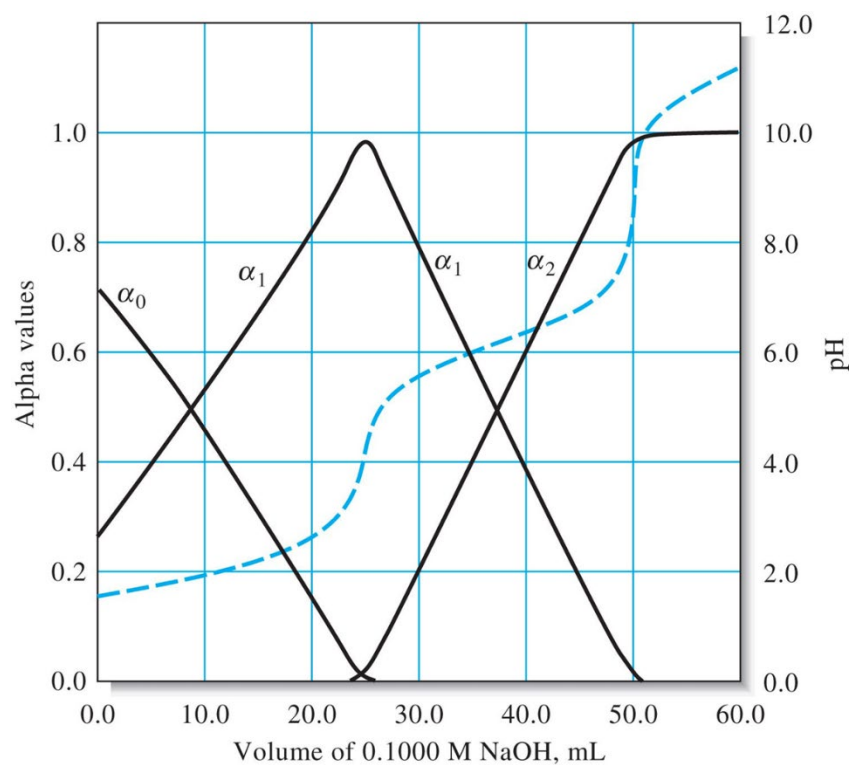
α_2 = 3rd term from the denominator

etc.

Alpha functions are useful for calculating the fraction of a species present at a given pH



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8. Calculate the fraction of PO_4^{3-} in a solution of H_3PO_4 where $\text{pH} = 9.35$.

H_3PO_4 $\text{pK}_{\text{a}1} = 2.15$

$\text{pK}_{\text{a}2} = 7.20$

$\text{pK}_{\text{a}3} = 12.30$

The effect of pH on solubility



$$K_{\text{sp}} = [\text{Ag}^+]^2[\text{CO}_3^{2-}]$$

We know that carbonate is a fairly strong conjugate base of a diprotic acid and that the concentration of carbonate will depend on the pH of the solution.



$$[\text{CO}_3^{2-}] = \alpha_2 C_{\text{T}}$$

(Go to next page for working example)

9. Calculate the solubility of Ag_2CO_3 at $\text{pH} = 4.00$. At a pH of 4.00 $\alpha_2 = 3.5 \times 10^{-9}$, and The K_{sp} for $\text{Ag}_2\text{CO}_3 = 8.10 \times 10^{-12}$

10. Calculate the molar solubility of CdCO_3 in aqueous solution that has a pH of 4.00. $\alpha_2 = 3.5 \times 10^{-9}$ $K_{\text{sp}} \text{CdCO}_3 = 1.8 \times 10^{-14}$